

FRI / FRII program architecture

1 INPUTS

Radio-galaxy images and class lists
Host-galaxy positions and source information

2 MORPHOLOGY AND MEASUREMENTS

Radio-emission extraction
greyscale, filtering,
thresholding,
connected regions and PCA

Ellipse description
centre, axes, orientation
and axial ratio

Skeleton and centreline
thinning, endpoint paths,
host-centred smoothing

3 PHYSICAL DATA AND SOURCE CHECKS

Radio flux and luminosity
calibrated survey images
centreline brightness profiles

Optical counterparts
and redshifts
black-hole and galaxy masses

Checking image geometry,
physical measurements,
optical data and redshift

4 OUTPUTS

Matched source measurements
missing values kept as missing

Centreline profiles, morphology
measurements, statistics
and clear images

SAMPLE-FILTERING AUDIT FRI / FRII

Initial
495 / 924

Visual removal
68 / 43

Automatic processing
not passed
44 / 297

Final accepted sample
379 / 547